

MENG 4205.002**LAB 7: INVESTIGATION OF COOLING RATE**

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ABSTRACT

This experiment investigates transient cooling behavior under forced convection using the TecQuipment TE93 Cross-Flow Heat Exchanger. The focus is to determine the rate of temperature decay of a heated copper rod as air passes across it at different valve openings. By plotting the logarithmic cooling curve, the convective heat-transfer coefficient (α) was obtained. This investigation uses Newton's law of cooling to quantify energy transfer from the rod surface to the moving air stream. Results show that higher airflow velocities significantly increase the rate of cooling, confirming that the convective coefficient rises with the Reynolds number. This study reinforces the principles of transient heat transfer and validates theoretical convection correlations.

Keywords: Cooling rate, transient conduction, heat-transfer coefficient, Nusselt number, Reynolds number, forced convection, cross-flow.

Nomenclature

α	Heat-transfer coefficient ($W/m^2 \cdot K$)
A_1	Effective surface area of heated rod (m^2)
m	Mass of heated rod (kg)
c	Specific heat of copper ($J/kg \cdot K$)
T_1	Ambient air temperature (K)
T_2	Heated rod temperature (K)
t	Cooling time (s)
M	Slope of cooling curve
ν	Kinematic viscosity (m^2/s)
Nu, Re, Pr	Nusselt, Reynolds, and Prandtl numbers
ρ	Density of air (kg/m^3)

1. INTRODUCTION

The cooling of heated bodies by moving air is a fundamental problem in thermal-fluid systems, particularly in heat exchangers and electronic cooling. When air flows across a hot surface, convective heat transfer removes energy, and the surface temperature decreases exponentially with time. This

phenomenon is described by Newton's law of cooling [1] [2]. The TE93 Cross-Flow Heat Exchanger enables measurement of this transient cooling rate by monitoring a heated copper rod as air passes through a rod array [3]. Understanding this cooling rate is essential for predicting system efficiency and designing equipment where heat must be dissipated effectively.

2. THEORY

The rate of heat loss from the rod follows Newton's law of cooling:

$$\dot{q} = \alpha A_1 (T_2 - T_1) \quad (1)$$

The energy balance for a small temperature change is expressed as:

$$\dot{q} dt = -m c dT \quad (2)$$

Combining and simplifying (1) and (2) gives:

$$dT / (T_2 - T_1) = -(\alpha A_1 / m c) dt \quad (3)$$

Integrating both sides between the limits of integration between 0 and t with respect to t yields:

$$\ln(T_2 - T_1) = -(\alpha A_1 / m c)t + \ln(T_0 - T_1) \quad (4)$$

Rewriting using base-10 logarithms:

$$\alpha = -2.3026 \times (m c / A_1) \times M \quad (5)$$

The slope M of the plot of $\log_{10}(T_2 - T_1)$ versus time represents the cooling rate, from which α can be determined. This relation confirms the exponential temperature decay predicted by Newton's law of cooling [1][4].

3. METHODS

3.1 Preparation

All connections on the TE93 apparatus were checked for tightness and leaks. The mass, dimensions, and surface area of the heated copper rod were measured. Ambient air temperature (T_1) and barometric pressure were recorded. The instrumentation panel zeroed before measurements.

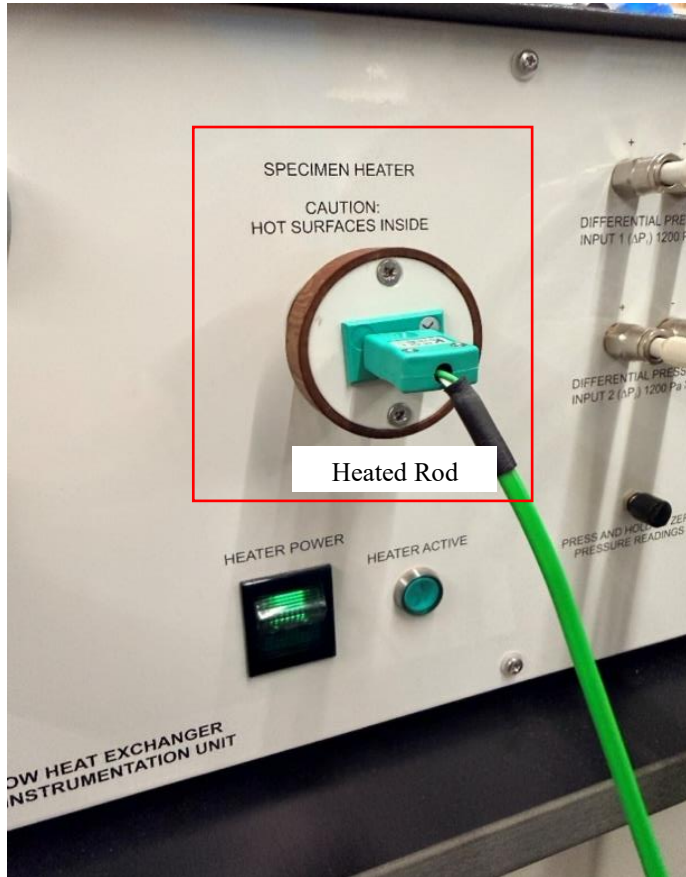


Figure 1 Heating Rod Inserted

3.2 Data Collection, Reduction, and Calculation

The copper rod was heated to approximately 75°C using the instrument's heater unit in Figure 1. Once stable, it was quickly transferred into the airflow duct, where it was exposed to air at a fixed valve setting. The temperature (T_2) was recorded every 10 seconds until the rod approached ambient temperature. The process was repeated for air-valve openings of 10% to 80% and for multiple rod positions (Columns 1–4), as well as a single-rod configuration. The temperature difference ($\Delta T = T_2 - T_1$) was recorded as seen in Figure 2, and $\log_{10}(\Delta T)$ was plotted against time. The slope (M) of this line was used in relation to determine α , see equation (5).

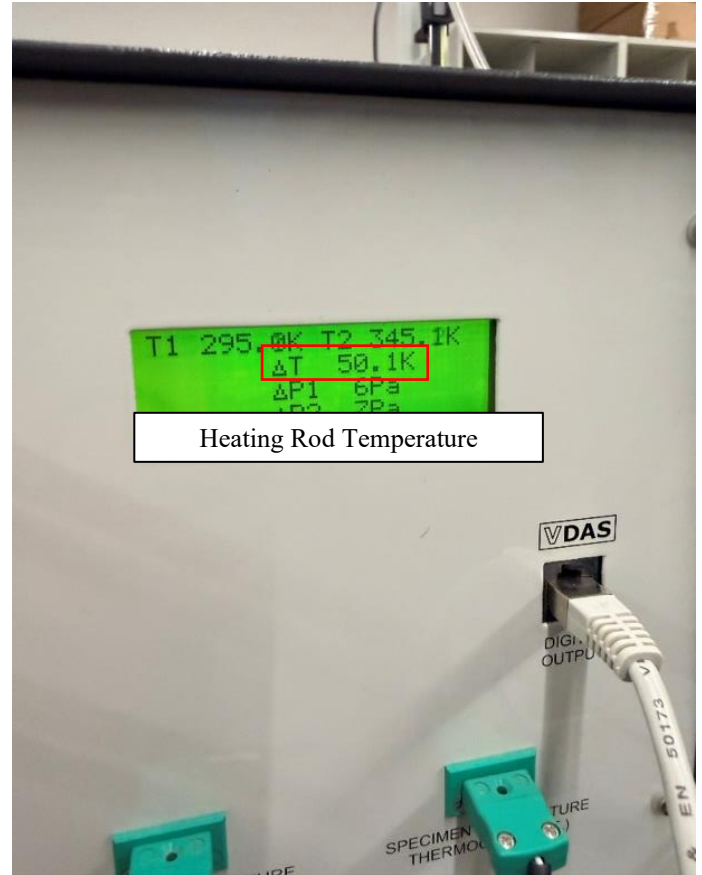


Figure 2 Display of the Heating Rod Temperature

4. EXPERIMENTS RESULTS AND DISCUSSION

4.1 Results

The cooling curves exhibited a linear relationship between $\log_{10}(\Delta T)$ and time, confirming exponential decay. At higher air-valve openings, the slope ($|M|$) increased, indicating faster cooling rates. This trend aligns with theoretical predictions, as higher air velocity enhances convective heat transfer. A representative sample of recorded data and plotted cooling curves are shown in Table 1.

Table 1 Sample cooling data for heated rod at 20% airflow with ambient temperature $T_1 = 294.9 \text{ K}$

Time (s)	T_2 (K)	ΔT (K)	$\log_{10}\Delta T$ (K)
0	344.1	49.2	1.7
10	341.1	45.6	1.7
20	338.2	42.5	1.6
30	335.3	40.4	1.6
40	332.8	37.9	1.6
50	330.3	35.4	1.5
60	328	33.1	1.5

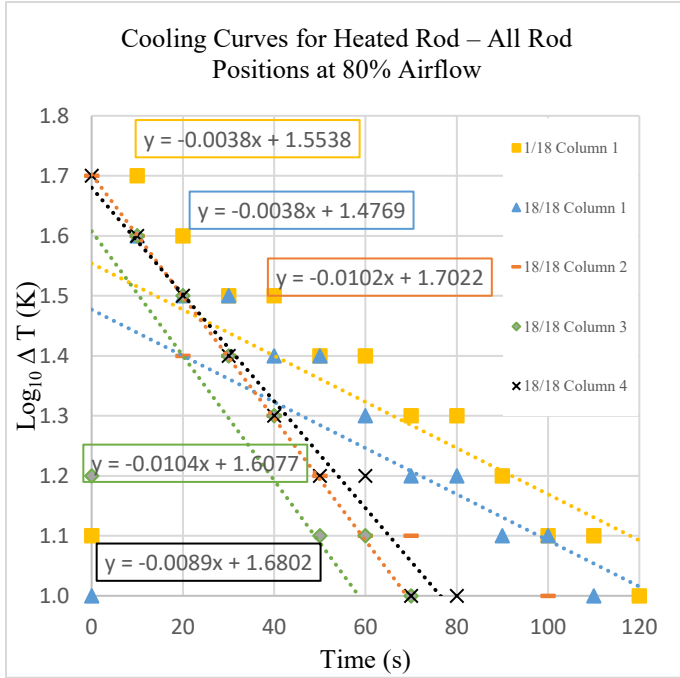


Figure 3 Cooling curves at 80% airflow for all rod positions. Column 3 cools fastest while the single-rod Column 1 cools slowest

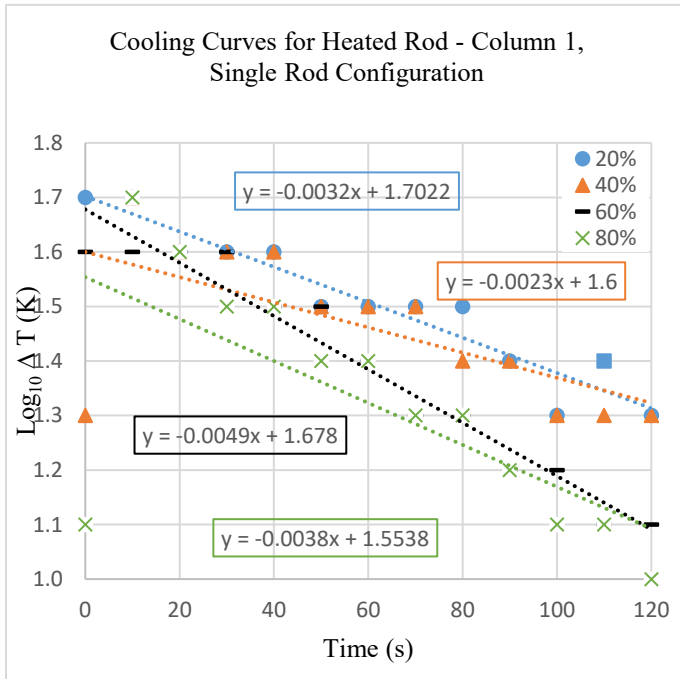


Figure 4 Cooling curves for single-rod configuration at 20-80% airflows.

Figure 3 demonstrates heated rod that was tested under five configurations to evaluate the influence of position and flow conditions on its cooling rate. Initially, the rod was placed alone in Column 1 (first configuration) to establish a baseline with the remaining holes filled with blanking plugs [3]. Afterwards, all rods were installed, and the heated rod was moved through Columns 1–4 to determine how its location within the array affected convective heat transfer (second configuration).

The results indicated that rod placement significantly influenced cooling behavior. The slopes of the logarithmic temperature difference curves at 80% airflow ranged approximately from -0.0038 s^{-1} in Column 1, -0.0102 s^{-1} in Column 2 and ultimately reaching a maximum of -0.0104 s^{-1} in Column 3, indicating the highest convective heat transfer coefficient. Column 4 also exhibited rapid cooling, though slightly less pronounced than Column 3.

In both configurations of Column 1 the slopes were nearly identical (-0.0038 s^{-1}) which shows that there is little effect from adjacent rods at the upstream position. However, downstream positions were much steeper, with column 3 cooling approximately 2.7x greater than Column 1 configurations.

The impact of airflow percentage of the single rod configuration can be seen in Figure 4. The slopes did not follow an increasing trend as airflow increased due to variability in the initial heated rod temperature across trials. Nonetheless, the rod position's effect shown in Figure 4 is consistent and reproducible. Rods located further back in the array cooled faster than those closest to the inlet.

4.2 Sample Calculations

Given: $m = 0.045 \text{ kg}$, $c = 380 \text{ J/kg} \cdot \text{K}$, $A_1 = 0.0040 \text{ m}^2$, slope $M = -0.0091 \text{ s}^{-1}$.

$$\alpha = -2.3026 (m c / A_1) M$$

$$\alpha = -2.3026 (0.045 \times 380 / 0.0040)(-0.0091)$$

$$\alpha = 90.0 \text{ W/m}^2 \cdot \text{K}$$

Hence, $\alpha \approx 90 \text{ W/m}^2 \cdot \text{K}$, which falls within the expected range for forced convection cooling of small cylindrical rods.

4.3 Ideal vs. Actual Comparison

To compare experimental and theoretical results, the ideal convective heat-transfer coefficient (α_{ideal}) was estimated using standard forced-convection correlations for flow over a cylinder. The theoretical Nusselt number is given by:

$$Nu = 0.24 Re^{0.6} Pr^{0.36} \quad (6)$$

The Reynolds and Prandtl numbers are respectively defined as:

$$Re = (\rho V D) / \mu \quad (7)$$

$$Pr = (c_p \mu) / k \quad (8)$$

Once Nu is known, the ideal convective coefficient can be calculated as:

$$\alpha_{(Ideal)} = (Nu k) / D \quad (9)$$

****Sample Ideal Calculation:****

Given: $V = 8.5 \text{ m/s}$, $D = 0.012 \text{ m}$, $\rho = 1.18 \text{ kg/m}^3$, $\mu = 1.85 \times 10^{-5} \text{ Pa}\cdot\text{s}$, $k = 0.026 \text{ W/m}\cdot\text{K}$, and $Pr = 0.71$,

$$Re = (1.18 \times 8.5 \times 0.012) / (1.85 \times 10^{-5}) = 6.51 \times 10^3$$

$$Nu = 0.24 (6.51 \times 10^3)^{0.6} (0.71)^{0.36} = 32.1$$

$$\alpha_{(Ideal)} = (32.1 \times 0.026) / 0.012 = 69.6 \text{ W/m}^2 \cdot \text{K}$$

Hence, the ideal convective heat-transfer coefficient is approximately $70 \text{ W/m}^2 \cdot \text{K}$ for this airflow condition.

Table 2 Ideal vs Actual α Comparison Table

Air Valve (%)	$\alpha_{(Ideal)}$ (W/m ² ·K)	$\alpha_{(Actual)}$ (W/m ² ·K)	Deviation (%)
20	46.6	31.5	32.4
40	66.3	22.6	65.9
60	86	48.2	44
80	91.2	37.4	59

4.4 Discussion

From the results, it was clear that the cooling rate increased as the airflow was raised. When the valve was opened more, the slopes of the logarithmic temperature curves became steeper, meaning the rod cooled faster. This occurred because higher air velocity improves convection by breaking up the thermal boundary layer and carrying away more heat from the surface [4].

Rod position in the array also made a noticeable difference. Rods that were placed further downstream, such as in Column 3 cooled the fastest, while those closer to the inlet cooled the slowest. These observations satisfy our physical intuition given that as air moves through the rods, it becomes more turbulent, and that turbulence helps improve mixing and overall heat transfer. The results followed the general convection trend predicted by (6), showing that the heat-transfer coefficient increases with Reynolds number.

Some differences between the experimental and theoretical α values likely occurred because the rod was not always heated to the exact same initial temperature in serial

trials. In addition, heat was lost to the surrounding air when transferring the rod from the heater to the airflow duct, which caused slight variations in the starting temperature. Uneven airflow inside the duct could also have contributed to error, since small variations in velocity across the cross-section may have caused certain areas of the rod to experience more or less cooling than others. This would lead to inconsistent temperature readings even at the same valve setting. Another factor considered was thermocouple lag, the delay that occurs when the temperature sensor takes time to match the actual surface temperature of the rod. This is because cooling acts quickly in this experiment, even a short response delay can make the recorded temperature slightly lower or higher than the true value at that moment.

Although these factors introduced minor variations, the overall trends still matched theoretical expectations; higher air velocity and turbulence resulted in faster cooling and a higher convective heat-transfer coefficient.

4.5 Conclusion

The TE93 Cross-Flow Heat Exchanger effectively demonstrated transient cooling and convective heat transfer principles. Measured temperature decay closely followed the exponential behavior predicted by Newton's law of cooling for each configuration.

Rod placement significantly played a major role in the rate of cooling and produced consistent results. As shown in *Figure 3* single rod configuration produced the slowest rate of cooling ($M = -0.0038 \text{ s}^{-1}$), due to minimal disruption of airflow through the column. Conversely, the all rods fitted Column 3 configuration produced the greatest cooling rate ($M = -0.0104 \text{ s}^{-1}$), which is a 2.7x fold increase in cooling rate relative to the single rod in Column 1.

The reason that the cooling rate varies with position of the rod with the array is due to the turbulence generated as the airflow passes through the rod array. Hence the rods located further back of the airflow experience more intense wake interaction and fluid mixing than those upfront, which results in a substantial increase in the convective heat transfer coefficient at these locations as seen in *Table 2*.

Although there were discrepancies in our initial temperatures for the heated rod which affected the ability to accurately compare different airflow percentages, it was still evident how the rod positioning affected cooling. Overall, the results support Newton's law of cooling and quantified the impact of turbulence on cooling performance in a forced convection system.

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In Depth Graphs -

